

Preliminary; please do not circulate

Modeling clean cooking transitions in Ghana: A cost-benefit analysis of LPG and electric scenarios using updated health and climate tools

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Over two billion people rely on solid, smoky cooking fuels which results in household air pollution (HAP) and subsequent devastating health outcomes, particularly for women and children, including stunting, cardiovascular disease, chronic obstructive pulmonary disease, and adverse pregnancy outcomes (1). Moreover, transitions to clean cooking fuels by 2040 could reduce carbon emissions by 15-32% (off a base of 1.02Gt-CO₂e) (2).

Despite the potential for health, climate, and social benefit, limited work has explored quantifying those benefits (3, 4), leaving policy makers without actionable evidence on clean fuel pathways to pursue or their potential benefits. A global analysis found that clean cooking transitions would provide net benefits of 1.4 trillion USD from 2020–2050 across 120 LMICs (5); however, this work does not include the costs of fuel infrastructure development, account for fuel stacking, or provide stakeholders insights into which clean fuels may be the most cost effective. Further, most studies quantifying the costs and benefits of clean cooking transitions do not rely on the most robust or recent advancements in health and climate modeling. Air Pollution Burden of Disease Explorer (ABODE) is currently the most robust tool for estimating health benefits from a reduction in exposure to particulate matter associated with a clean cooking transition (6), while our team adjusted the FaIR climate model to assess climate benefit from clean cooking transitions (2).

Even ABODE and FaIR, when used to characterize numerous countries and regions, require a number of literature-based assumptions or regional estimates that may not be relevant or appropriate for stakeholders when creating national policy and planning. The lack of specificity in cost benefit analyses as well as health and climate modeling limit the rigor of the estimated net benefits or health/climate impact.

In collaboration with Ghana's Ministry of Energy, we propose to update FAIR and ABODE with Ghana specific parameters and estimate more robust and relevant cost-benefit analyses of specific policy-relevant clean fuel pathways for stakeholders.

Our team will build four different clean fuel scenarios: a business-as-usual, a policy transition, a full transition to LPG, and a full transition to electric cooking. To investigate the scenarios with the most robust evidence to date, we will update ABODE with personal exposure measurements from the GRAPHIS study, include sub-national ambient air pollution estimates, and update the population data, the solid fuel use estimates, and the GDP per capita. We will update the FAIR model with Ghana specific parameters, such as LPG emission factors, Ghana's electric grid mix, district level fraction of non-renewable biomass estimates from MoFuSS, a dynamic, spatial-temporal landscape model (7), stove emission factors, etc. We will further update other costs and benefits with Ghana specific estimates, such as infrastructure costs, and gender benefits and time estimates.

This work will provide policy makers in Ghana with robust evidence to motivate further investment into clean cooking transitions, and clarity on where to invest. We can build upon this work to investigate further where each fuel may have the most benefit or apply this type of analysis to provide evidence for another country's planning. Clear evidence will be critical for policymakers and stakeholders to make progress countries towards clean fuel transitions. This document provides a brief overview of our methods and initial results.

Methods

Scenarios

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We develop four key scenarios. Our primary scenario for comparison is a business-as-usual (BAU) scenario, which we develop by using the WHO's household cooking fuel data disaggregated into rural and urban populations for Ghana and assume that future household fuel use follows historical trends (2000-2020) (8). We compare BAU to three alternate scenarios (outlined in Figure 1):

- (1) Policy Transition (IT) scenario in which fuel adoption trajectories meet specific policy targets for biogas, improved charcoal stoves, ethanol, LPG, and electric cooking for urban and rural populations (see Table 1). Polluting fuels reduce accordingly and equally

Table 1. Policy Transition fuel targets by fuel, locality, and year

Fuel	Locality	2030	2040	2060
Biogas	rural	0.3%	1.0%	0.2%
Biogas	urban	0.7%	1.0%	0.8%
Electric	rural	1.0%	5.0%	8.0%
Electric	urban	7.0%	25.0%	50.0%
Ethanol	rural	1.0%	1.0%	2.0%
Ethanol	urban	1.0%	3.0%	4.0%
Improved charcoal	rural	14.0%	10.0%	10.0%
Improved charcoal	urban	25.0%	15.0%	5.0%
LPG	rural	10.0%	14.0%	15.0%
LPG	urban	40.0%	25.0%	5.0%

- (2) Full Transition (FT) to LPG scenario in which household fuel adoption trajectories shift dramatically and reach universal use of LPG by 2060. The FT scenario follows a trajectory in which LPG completely replaces polluting fuels (wood, charcoal, and kerosene) by 2060. We assume other clean fuels like electricity maintain their BAU trajectories.
- (3) FT to electric cooking scenario in which household fuel adoption trajectories shift dramatically and reach universal use of electricity for cooking by 2060. The FT scenario follows a trajectory in which electric cooking completely replaces polluting fuels (wood, charcoal, and kerosene) and even the baseline LPG use by 2060.

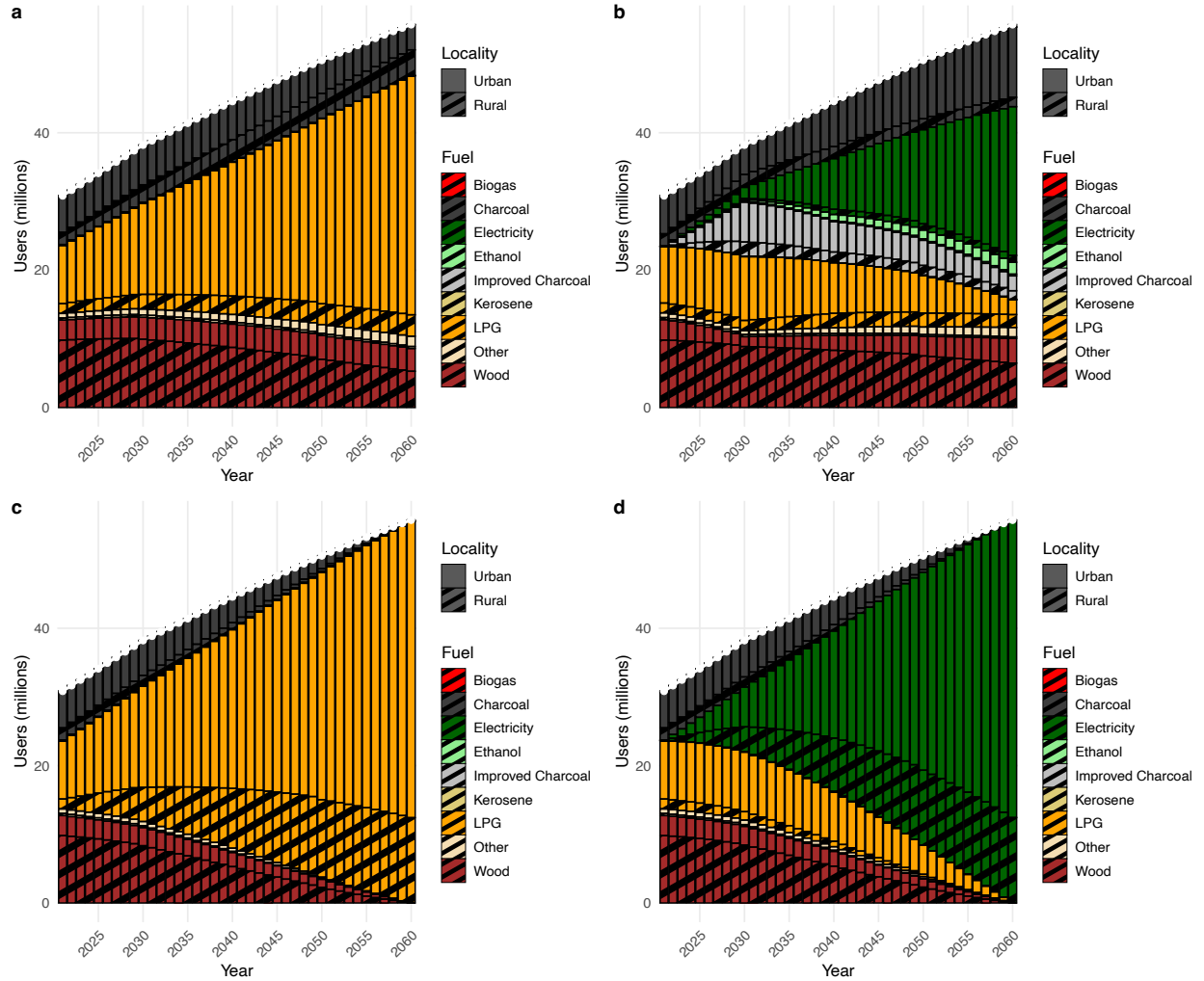


Figure 1. Clean cooking scenarios for Ghana 2021-2060. We depict the number of users by fuel type and locality in each of our four transitions: business-as-usual (panel a), policy transition (panel b), full transition to LPG (panel c), and full transition to electric cooking (panel d). The white line indicates total Ghanaian population for each year between 2021-2060.

Using UN population projections, we estimate Ghana's population at the district level and urban/rural setting between 2021 to 2060, ensuring alignment with the 2021 census. We apply the percentage of users using each fuel to calculate the total number of users of each fuel. We then assume 2 MJ-delivered / (capita day) and determine total annual fuel demand for each scenario.

Using this annual fuel consumption, we then estimate CO_2 , CH_4 , N_2O , SO_x , CO , non-methane volatile organic compounds (NMVOCs), NO_x , BC, and OC using emission factors from field studies for firewood and charcoal (9–11) and the GREET model (12) with Ghana specific parameters for LPG and electricity. We also use Ghana and locality specific parameters for fNRB settings from the MoFuSS.